

**INVESTIGATING THE IMPACT OF FIRM-GENERATED CONTENT ON
USER-GENERATED CONTENT AND BOX OFFICE SALES**

Basil Kazi

An honors thesis submitted to the faculty
of the Kenan-Flagler Business School at the
University of North Carolina at Chapel Hill

Chapel Hill
2016

Approved by _____
(Dr. Shijie Lu)

ABSTRACT

Basil Kazi

Investigating the Impact of Firm-generated Contents on
User-generated Contents and Box Office Sales
(Under the direction of Dr. Shijie Lu)

This thesis examines how firm-generated content in the movie industry affects user-generated content, which in turn affects box-office sales. This study first examines how user-generated content affects box-office sales. Next, the relationship between firm-generated content and user-generated content is analyzed. To achieve these objectives, I collect the data of 224 movies over a two-year period as well as the associated tweets generated by both users and firms, the latter of which include the movie studio and main actors and actresses. My results indicate that the increased volume of Tweets drives box-office sales. I also find that the volume of firm-generated content, either generated by movie studios or by actors, results in an increase in the volume of user-generated content. In addition, I also find that interactions between official movie accounts and actor accounts on Twitter produces a synergy that boosts the volume of user-generated content. Several managerial implications regarding the effective use of social media as a communication channel in the movie industry are discussed.

TABLE OF CONTENTS

ABSTRACT.....	ii
LIST OF TABLES.....	v
LIST OF FIGURES.....	vi
Chapter	
I. INTRODUCTION.....	1
II. LITERATURE REVIEW.....	2
A. User-generated content.....	3
B. Firm-generated content.....	5
C. Employee-generated content.....	8
D. Movies and online word of mouth.....	8
III. METHODOLOGY.....	10
A. Data description.....	10
1. Movies.....	10
2. Tweets.....	12
B. Econometric methods.....	13
1. User-generated content's relationship to box-office sales.....	13
2. Firm-generated content's relationship to user-generated content.....	15
IV. PRELIMINARY ANALYSIS.....	17
A. Volume of user-generated content and its relationship with release date.....	17
B. Valence of user-generated content and its relationship with release date.....	18

C.	Volume of firm-generated content in relation to release date.....	20
V.	RESULTS.....	22
A.	User-generated content's relationship with box-office revenue.....	22
B.	Firm-generated content's relationship with user-generated content.....	24
1.	Movies.....	24
2.	Actors.....	27
3.	Movies and Actors.....	28
VI.	CONCLUSION.....	30
VII.	APPENDIX	
A.	Summary statistics of movie sample.....	33
B.	Breakdown of movie account tweets	34
C.	Breakdown of actor account tweets	35
D.	Breakdown of user account tweets	36
VII.	REFERENCES.....	37

LIST OF TABLES

TABLE 5.1.....	22
TABLE 5.2.....	25
TABLE 5.3.....	27
TABLE 5.4.....	29

LIST OF FIGURES

FIGURE 4.1.....	17
FIGURE 4.2.....	20
FIGURE 4.3.....	20

I. INTRODUCTION

By 2019, online advertisement will become the largest segment of film industry advertisement expense. At the same time, entertainment film revenues will also hit an all-time high of \$104.62 billion dollars (PricewaterhouseCoopers, 2015). Due to the increasing size of the entertainment film market and Internet advertising's increasing prevalence, movie production companies must be able to adapt their marketing tactics to communicate appealing messages to potential moviegoers. Due to the spread of social media, marketers can now move beyond simple push advertisement, where users are forced to watch trailers or have banner advertisements taking up part of their screen. Instead, marketers can now directly interact with potential moviegoers by entering the realm of word-of-mouth marketing, where they can influence the conversation that sways potential moviegoers.

If firm-generated online content has positive and significant effects on user-generated content about a particular film, then most film production companies would be active on social media. The opposite can be said if firm-generated content had little positive effect or even a negative effect on user-generated content. Then why do some major movies have official Twitter accounts and others do not? Do official movie accounts' Tweets even make a difference on firm's revenues? What about Tweets from actors' personal Twitter accounts? Is active social media participation worth the time and money, or could it be counterproductive, simply bothering individuals who do not go on social media to be pressed with advertisements?

This research aims to identify the effects of firm-generated content on user-generated content and box-office sales. The understanding of these effects can provide

new insights on how to leverage online social media to increase ticket sales and to improve user experience with firm-generated content. By examining firm-generated content, user generated content and box-office sales for movies released between July 2013 and June 2015, I quantify firm-generated content's effects on user-generated content and user-generated content's effects on box-office sales. Specifically, the first portion of this research establishes user-generated content's effects on box-office sales, while the second portion examines firm-generated content's effects on user-generated content. I find that the volume of firm-generated content drives user-generated content and the volume of user-generated content drives box-office sales. Interestingly, I also find that the interaction between official movie accounts and personal actors' accounts can produce a synergy that drives the volume of user-generated content.

II. LITERATURE REVIEW

Research analyzing social media and its role in word-of-mouth marketing is extensive, yet primarily focuses on user-generated content. Within this literature, researchers commonly use movies and box office sales in order to quantify the effect of user-generated content.

I have divided this literature review into four sections. Initially, I examine research that focuses on word-of-mouth marketing and online content. Therefore, the first three sections of my literature review will be divided into:

- (1) user-generated content
- (2) firm-generated content
- (3) employee-generated content

In the fourth and final section, I examine research to justify the rationale behind my selection of movies as a sample, while I research firm-generated content's effects on user-generated content.

User-generated content

The vast majority of research analyzing social media and its role in word-of-mouth marketing focuses on user-generated content. This focus on user-generated content can largely be attributed to the links between the formation of user-generated content and customer action, specifically user-generated content's ability to drive and predict customer action. (Shriver, Nair, & Hofstetter, 2013). Furthermore, user-generated content presents firms with the opportunity to take advantage of network effects. Researchers studying user-generated content tend to focus on the same variables when studying online content and often use these variables to measure user-generated content's predictive strength.

Variables to consider when examining online content

Within the literature examining the impact of user-generated content, the most common variables measured are:

- (i.) volume, the number of user-generated reviews
- (ii.) valence, the proportion of user-generated reviews that are positive, negative, or neutral
- (iii.) variance, the spread among user-generated reviews

When looking at sequential rollout and aggregation of movies across local markets, valence is the primary indicator of a movie's success. However, nationally, volume is the primary indicator of a movie's success (Chintagunta, Gopinath & Venkataraman, 2010). Furthermore, negative reviews have a more powerful impact than positive reviews on user-generated content (Mayzlin, 2006), a quality that transcends industries, as I will examine in the next section.

User-generated content as a predictive measure

Due to user-generated content's association with customer action, user-generated content has been examined as a predictive measure for stock market performance. For example, Tirunillai and Tellis (2012) investigated whether or not user-generated content could be used to predict stock market performance and, if so, what the strongest predictive measures were. In their study, an increase in the volume of chatter indicated positive returns for the few days after the increase in chatter. Positive user-generated content indicated no significant effects on stock performance; however, negative user-generated content predicted statistically significant negative returns. Furthermore, an increase in volume of user-generated content regarding a firm was found to negatively

influence competitive firms' stock prices (Tirunillai & Tellis, 2012). Other studies, some of which focused specifically on user-generated content's predictive strength on movie success, produced similar takeaways (Asur & Huberman, 2010; Tirunillai & Tellis, 2012; Wong & Chiang, 2012).

Asur and Huberman (2010) used Twitter posts, or Tweets, to quantify online chatter. Their research used regression analysis on 2.89 million Tweets for 24 movies and found remarkable success using the Tweet-rate (Tweets per hour) and a PN ratio (positive Tweets to negative Tweets) as their only variables to predict box-office sales. The regression model Asur and Huberman created using just Twitter data was more accurate than the Hollywood Stock Exchange, an information market regularly used to predict box-office sales. With the understanding that positive user-generated content drives ticket sales, movie studios stand to benefit from the ability to boost positive user-generated content (Chintagunta, Gopinath & Venkataraman, 2010).

Firm-generated content

One way to boost positive user-generated content could be to manage firm-generated content. Unlike user-generated content, firm-generated content is primarily positive, since the firm has control of the content and presumably prefers a positive message. Literature on firm-generated content focuses on how firm-generated content impacts user-generated content, how firm-generated content impacts customer behavior, and how users interact with firms.

How firm-generated content impacts user-generated content

Few studies have examined firm-generated content's relationship to user-generated content and none, to my knowledge, have examined firm-generated content's

relationship to user-generated content within the entertainment industry. In 2013, Miller and Tucker took an in-depth look at user-generated content in the health care industry. In their research, Miller and Tucker discovered that firm-generated content would drive other online content; however, that content primarily consists of employee-generated content. This discovery is interesting, because it suggests that social media posts by a firm may not result in consumer engagement, even if overall online engagement increases. Furthermore, Miller and Tucker argue that firm-generated content could influence potential customers if firm posts are primarily focused on customer experience. However, most posts do not seem to be targeting customers' interests, instead focusing on the firm's direction and accomplishments.

Miller and Tucker's study focused on the health care industry, which has limitations when trying to extrapolate insight to other industries. For example, the health care industry's use of social media is probably not representative of most industries, due to the United States' citizens' restrictions in choosing their hospitals (Miller & Tucker, 2013). Furthermore, an individual's health care experiences are less likely to be shared since the health care industry has comparatively very high privacy concerns (Goldfarb & Tucker, 2012).

In 2013, Risika, Kumar, Janakiraman and Bezawada identified "customers who exhibit a strong patronage with the firm, buy premium products, and exhibit lower levels of buying focus and deal sensitivity" as the customers on whom firm-generated content had the strongest effect (p. 108). The same researchers along with Kannan identified "experienced, tech-savvy, and social media prone customers" (Kumar, Bezawada,

Rishika, Janakiraman & Kannan, 2015, p. 2) as customers most likely influenced to buy products and services by firm-generated content.

How firm-generated content impacts customer behavior

In 2015, Kumar et al. examined a physical store for a focal retailer, this store's online firm-generated content, and this content's effects on customer behavior, concluding that firm-generated content has a powerful effect on customer behavior, even when controlling for television advertisement and email marketing. Out of the three components of firm-generated content that were measured: receptivity, valence, and customer susceptibility, receptivity had the most positive and significant impact. Receptivity, in this study, is defined as the amount of firm-generated content taken in. Valence is how positive or negative firm-generated content is and customer susceptibility is defined as how easily a customer is influenced by firm-generated content. Although weaker than receptivity, valence and customer susceptibility also had positive and statistically significant impacts (Kumar et al., 2015). Although this paper looked into firm-generated content in social media and its relationship to customer spending, customer cross-buying, and customer profitability, it left questions regarding firm-generated content as an avenue to impact user-generated content as an area for future studies.

How users interact with firms

Shahbaznezhad and Tripathy (2015) categorized user behavior and engagement with firm-generated content as "creative attitude" and "following attitude." "Creative attitude" captures content creation like creating a new hashtag or replying, while "following attitude" captures agreement, like sharing or retweeting a post. It is worth

noting that social media experts argue that customers should view a firm's social media presence as organic for optimal results (Gossieaux & Moran, 2010).

Employee-generated content

As discussed earlier, Miller and Tucker's (2013) examination of firm-generated content revealed that firm-generated content could drive employee-generated content. However, there is no study to my knowledge that examines employee-generated content's effect on user-generated content. The effects of employee-generated content on user-generated content are particularly important in the movie industry, where actors have a significant effect on movie revenues (Liu, Mazumdar & Li, 2014).

Due to the significant investment studios put into movie stars, Liu, Mazumdar and Li (2014) investigated the amount of revenue movie stars bring in versus non-stars. The researchers did so by developing an endogenous switching regression model, in which they compare movies in which a star acted in the movie versus if a nonstar had acted in the same movie and vice-versa. The differences in revenue an actor can bring to a movie can be attributed to (1) how an actor can affect movie characteristics and (2) how an actor is able to draw in returns without affecting movie characteristics, like effecting media coverage which would drive sales.¹

Movies and online word-of-mouth

Researchers examining online word-of-mouth's effects on sales often select movies to appear in their academic writing since (1) movies take up a reasonable amount of chatter in social media communities (Wong, Sen, Chiang, 2010) and (2) outcomes of movie "success" can easily be judged by looking at the box-office sales (Asur &

¹ Since actors, whether or not they are stars, affect movie sales, one could question how these actors interactions with potential moviegoers on social media translate to sales.

Huberman, 2010). I used movies in this study for these two reasons and to contribute to the literature of word-of-mouth's effects on sales.

Research regarding word-of-mouth in social media primarily focuses on user-generated content and its effects on market outcomes. However, despite the established effects of user-generated content and market outcomes, the literature does not directly address how firms can affect user-generated content in a way that positively impacts market outcomes. This research will attempt to fill this void by exploring and analyzing how firm-generated content affects user-generated content and box-office sales.

III. METHODOLOGY

To examine the possible effects of firm-generated content on user-generated content and ultimately box-office sales, this study will first establish user-generated content's relationship with box-office sales. After establishing which factors influence user-generated content's relationship with box-office sales, this study focuses on examining how firm-generated content affects these factors. This chapter will include (1) data description and (2) econometric methods.

Data description

Movie data and Twitter data were collected to examine how user-generated Tweets might affect box-office sales and how firm-generated Tweets might affect user-generated Tweets. In this section, I explain which movies and Tweets I selected and which of their characteristics were examined.

Movies:

Movies were specifically selected to study the relationship of firm-generated content, user-generated content and revenue because: (1) movies are frequently discussed online and (2) movie success is easy to measure through publically available box-office sales (Wong, Sen & Chiang, 2010).

For this study, movies released between July 01, 2013 and June 30, 2015 were examined. Two years of data allowed seasonal and annual factors that potentially affected box-office sales to be controlled for. Recent movies were selected to properly examine user-generated online content, which has only established itself within the last several years. Movies within these dates were deleted from the sample if they:

1. Were documentaries, in order to focus on actors' influences when examining firm-generated content.
2. Were foreign or were not primarily English, in order to control for variations amongst international movies, such as their varying distribution networks and marketing tactics.
3. Had box-office revenue under \$1,000,000, which was an arbitrary success point selected in order to eliminate movies whose information would not be publically available.

To be able to match movies to the Twitter data, movies without an official Twitter account and hashtag were also eliminated from the dataset. The final dataset contained information for 224 movies. For each movie, the following measures of success were recorded to isolate the effect user-generated content has on these movies' box-office sales:

- Box-office sales for the first eight weeks since opening
- The movie budget
- IMDb ratings
- MPAA ratings
- The genre
- If any main actor ever had an Oscar nomination
- If the primary director ever had an Oscar nomination

The budget of the movie was collected to control for the influence the amount of money spent on a movie had on its eventual box office returns. IMDb ratings were collected to capture movie critics' opinions on the quality of the movie. Indicator

variables were used to factor in genre and MPAA ratings. For simplicity, genres were broken down into five categories and MPAA ratings into three categories. Also, binary variables were used to capture actor and director prestige. All data was collected from BoxOfficeMojo.com, IMDB.com and The-Numbers.com.

Tweets

Twitter was selected as the avenue to collect user-generated content due to its popularity and due to the fact that most accounts are publically accessible. Twitter is a social media site where individuals communicate through Tweets. Simply put, Tweets are messages posted on Twitter limited to 140 characters of text. Publically available Tweets can be *Liked* and *Retweeted* to show support. A “like” simply shows support, while a Retweet is a re-posting of someone else’s Tweet. Inserting a *Hashtag* to a Tweet marks the topic of that Tweet and allows users to categorize their Tweets.

Actor and movie account information were manually collected. Twitter verifies key individuals and brands on Twitter. For unverified accounts, I thoroughly examined Tweets in order to determine whether or not an account was authentic. Then, I submitted Twitter accounts and official hashtags were submitted to a freelancer who collected the firm-generated and user-generated Tweets mentioned immediately below.

Public user-generated Tweets were gathered to analyze their effects on box-office sales. To differentiate user-generated content about a specific movie from all other Tweets, all user-generated Tweets gathered included the official Twitter hashtag of the movie it refers too. Data for user-generated Tweets posted between January 01, 2013 and August 31, 2015 were gathered. For each date, the number of positive, neutral, negative and total Tweets were collected for each movie’s unique Twitter hashtag. In total,

17,014,979 user-generated Tweets were examined and 137,312 firm-generated Tweets were gathered. All Tweets and Twitter data were gathered and classified by a freelancer.

Hashtags were used to differentiate user-generated Tweets about a particular movie from other conversation on Twitter. Simply using movie names was not deemed appropriate because using generic names may draw out Tweets not referring to a movie. For example, collecting Tweets simply containing the movie name “Spy” would pull out Tweets such as “I think my neighbor’s a spy,” while using the #SpyMovie hashtag to collect would pull out Tweets like “Haven’t laughed so much in a while #SpyMovie.”

Firm-generated Tweets were gathered to analyze their effects on user-generated content. If any of the three main actors or the movie itself had a Twitter account, Tweets were pulled from these accounts. All firm-generated Tweets between the dates January 01, 2013 and August 31, 2015 were collected. Each Tweet’s date, content, number of retweets and number of likes were recorded. Each Tweet was also classified as normal, a reply or a retweet.

Econometric methods

As mentioned in the introduction of this section, this research examines the possible relationship between user-generated content and box-office sales prior to examining firm-generated content and its relationship to user-generated content. As such, this subsection is divided into examining: (1) online user-generated content’s relationship to box-office sales (2) online firm-generated content’s relationship to user-generated content.

User-generated content's relationship to box-office sales:

To isolate the impact of user-generated Tweets on box office sales, other potential drivers of sales were collected and tested through regression analysis along with the volume and valence of user-generated Tweets. Equation 1 will be produced by multiple linear regression analysis:

$$\ln(Rev)_{i,t+1} = \alpha_0 + \alpha_1 \ln(UGC)_{i,t} + \alpha_2 \left(\frac{Pos}{UGC} \right)_{i,t} + \alpha_3 \left(\frac{Neg}{UGC} \right)_{i,t} + \alpha_4 \ln(budg)_i + \alpha_5 Dram_i + \alpha_6 Horr_i + \alpha_7 Act_i + \alpha_8 Dir_i + \alpha_9 Act_i + \alpha_{10} GPG_i + \alpha_{11} PG13_i + \alpha_{12} IMDb_i + \alpha_{13} t_i + \varepsilon \quad (1)$$

The volume of user-generated Tweets (*UGC*) will be calculated by subtotalling the total number of user-generated Tweets in the week prior to the box-office revenue (*Rev*) it was aligned with. Both the box office sales and the volume of user-generated Tweets were inserted into a natural log function in order to counteract diminishing returns. Valence will be analyzed by examining two ratios: (1) number of positive Tweets (*Pos*) to total Tweets per movie per week and (2) number of negative Tweets (*Neg*) to total Tweets per movie per week. Like the volume, the valence of a particular week's Tweets will also be tied to the box-office sales for its respective movie one week after. A lag period between user-generated Tweets and box-office sales is used in order to analyze how user-generated content drives box-office sales, instead of showing that the two are simply correlated. Since information on social media spreads quickly, and since box-office sales are reported as weekly figures, a one-week lag period was selected because it was the smallest amount of possible lag that could be given to user-generated content. A variable for the week (*t*) of user-generated content will also be used properly reflect and capture the effect of time after release on box-office sales. The other potential drivers of sales for each movie include budget (*budg*), genre (*Dram*, *Comd*, *Horr* & *Acti*), MPAA

rating (*GPG* & *PG13*), IMDb rating, (*IMDb*) and whether or not the director and main actors have been nominated for an Oscar (*Dir* & *Act*). These other potential drivers are all independent of time.

Firm-generated content relationship to user-generated content:

Similar to the procedure that will measure user-generated content's effects on box-office sales, regression analysis will be used to isolate firm-generated content's impact on user-generated content. Like the earlier analysis, budget, genre, MPAA rating, IMDb rating, and whether or not the director and actor have been nominated for an Oscar will all be tested alongside the volume of firm-generated content. Unlike the earlier analysis, valence of firm-generated content will not be examined since it can be assumed that all firm-generated content positively reflects the movie it attempts to promote. Firm-generated content will be divided into movie accounts and actor accounts. To determine whether or not firm-generated content drives user-generated content, firm-generated volume will be subtotaled daily and paired with the volume of user-generated content one day later. Using the same rationale as the earlier analysis, a lag period between firm-generated content and user-generated content is used to analyze how firm-generated content drives user-generated content, instead of showing that the two are simply correlated. In this analysis, there will be a one day lag, the smallest lag possible with the dataset compiled. There will be three regression models run to measure the impact of official movie account Tweets, using regression analysis to formulate the following equations:

$$\ln(UGC)_{i,t+1} = \beta_0 + \beta_1 FGC_{i,t} + \beta_2 \ln(budg)_i + \beta_3 Dram_i + \beta_4 Comd_i + \beta_5 Horr_i + \beta_6 Act_i + \beta_7 Dir_i + \beta_8 Act_i + \beta_9 GPG_i + \beta_{10} PG13_i + \beta_{11} IMDb_i + \beta_{12} t_i + \varepsilon \quad (2)$$

Equation 2 will be used to measure official movie account Tweets (*FGC*) impact on user-generated content (*UGC*).

$$\ln(UGC)_{i,t+1} = \gamma_0 + \gamma_1 AGC_{i,t} + \gamma_2 \ln(budg)_i + \gamma_3 Dram_i + \gamma_4 Comd_i + \gamma_5 Horr_i + \gamma_6 Act_i + \gamma_7 Dir_i + \gamma_8 Act_i + \gamma_9 GPG_i + \gamma_{10} PG13_i + \gamma_{11} IMDb_i + \gamma_{12} t_i + \varepsilon \quad (3)$$

Equation 3 will be used to measure personal actor account Tweets (*AGC*) impact on user-generated content (*UGC*).

$$UGC_{i,t+1} = \delta_0 + \delta_1 FGC_{i,t} + \delta_2 AGC_{i,t} + \delta_3 FGC_{i,t} * AGC_{i,t} + \delta_4 \ln(budg)_i + \delta_5 Dram_i + \delta_6 Comd_i + \delta_7 Horr_i + \delta_8 Act_i + \delta_9 Dir_i + \delta_{10} Act_i + \delta_{11} GPG_i + \delta_{12} PG13_i + \delta_{13} IMDb_i + \delta_{14} t_i + \varepsilon \quad (4)$$

Equation 4 will be used in order to measure how movie accounts and actor accounts interactions (*FGC * AGC*) affect user-generated content (*UGC*).

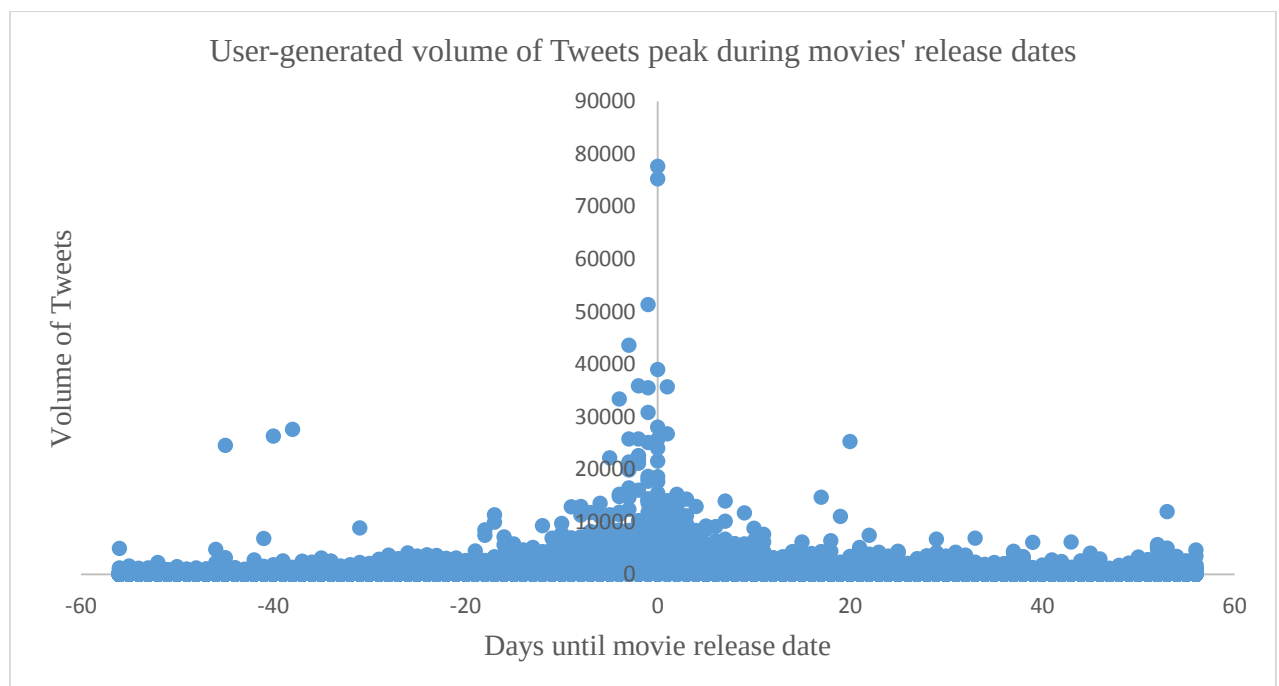
IV. PRELIMINARY ANALYSIS

For the purpose of comprehending user- and firm-generated content and to fully establish the data used in this study is well distributed and robust, user- and firm-generated content and their relationship with their movie's respective release dates were examined.

Volume of user-generated content and its relationship with release date

To understand how user-generated Tweets change over time, both the volume and valence of selected Tweets were examined in relation to time. Specifically, I examined user-generated Tweets 8 weeks prior to a movie's release until 8 weeks after a movie's release. Figure A, a time-series plot examining the volume of Tweets regarding a particular movie relative to a movie's release date, shows that the volume of Tweets appear to peak at a movie's release date and suggests a quadratic function would better indicate the relationship of user-generated content and time.

Figure 4.1



Similarly, a simple linear regression between volume and time did not yield significant results at a .05 level, indicating a linear relationship did not adequately capture the relationship between time and user-generated content. Furthermore, a quadratic regression between the volume of user-generated Tweets and their distance from the release date of the movie being discussed resulted in a significant negative coefficient, indicating that there is less user-generated content further away from a movie's release date. Conversely, it can be deduced that more user-generated content is produced closer to a movie's release date. When comparing changes in the volume of user-generated content prior to a movie's release to after a movie's release date, linear regression shows that the volume of Tweets increases prior to a movie's release faster than it decreases after a movie's release. The fact that the volume of Tweets decrease relatively slowly after a movie's release makes intuitive sense – new moviegoers can watch and join the conversation regarding the movie any time after its release date.

Valence of user-generated content and its relationship with release date

As discussed frequently in the literature, the valence of user-generated content, like the volume of user-generated content, plays a strong role in successful word of mouth marketing. To properly examine the valence of user-generated Tweets, percentages were used to analyze changes in positive, negative and neutral Tweets 8 weeks prior to a movie's release date to 8 weeks after. For example, more positive Tweets could mean the conversation was becoming more positive, but could also simply be indicative of an increase in the volume of Tweets in general. Using percentages controlled for the effects of changing volumes.

The percentage of positive Tweets in one day increased as time went on after a movie's release date, indicating positive conversation became more common as more individuals became more informed about a particular movie. Interestingly, the percentage of negative Tweets increased over time, as well. An increasing positive percentage of Tweets can only be paired with an increasing negative percentage of Tweets if the percentage of neutral Tweets decreased over time, which time-series analysis confirmed that it did. Examining the intercepts and slopes of the three percentages produced by regression analysis showed that neutral Tweets take up the largest percentage of Tweets during a movie's release date, followed by positive Tweets and then negative Tweets. The changes of the percentages over time signals conversation becomes more polarized as more individuals learn about and watch a particular movie. Moviegoers go from discussing information such as their plans to watch the movie to sharing their opinion on the quality of the movie. These changes in volume relative to time make intuitive sense - prior to a movie's release date, moviegoers are unfamiliar with a movie and rarely have strong opinions to share publically about the movie they will be watching. However, after a particular movie's release, new moviegoers are continuously watching the movie, and new opinions are being formed. An examination of the positive Tweet to negative Tweet ratio shows that negative Tweets increase faster than positive Tweets.² This being said, on average, the number of negative Tweets on any given day never surpasses the total amount of positive Tweets.

Considering the vast interest in user-generated content by firms, it is no surprise firm-generated content tries to influence the conversation into something more

² The same conclusion can be deduced by analyzing the slopes produced by the regression analysis of the positive Tweet percentage and negative Tweet percentage.

voluminous and positive. Therefore, in the next section, I will discuss my preliminary analysis for firm-generated content.

Volume of firm-generated content in relation to release date

To compare firm-generated content with user-generated content, both official movie account Tweets and their actors' personal account Tweets were examined in relation to time. A time series graph showing distance from release date and Tweets per day for both official movie accounts and personal actor accounts are inserted below as Figure B and C, respectively.

Figure 4.2

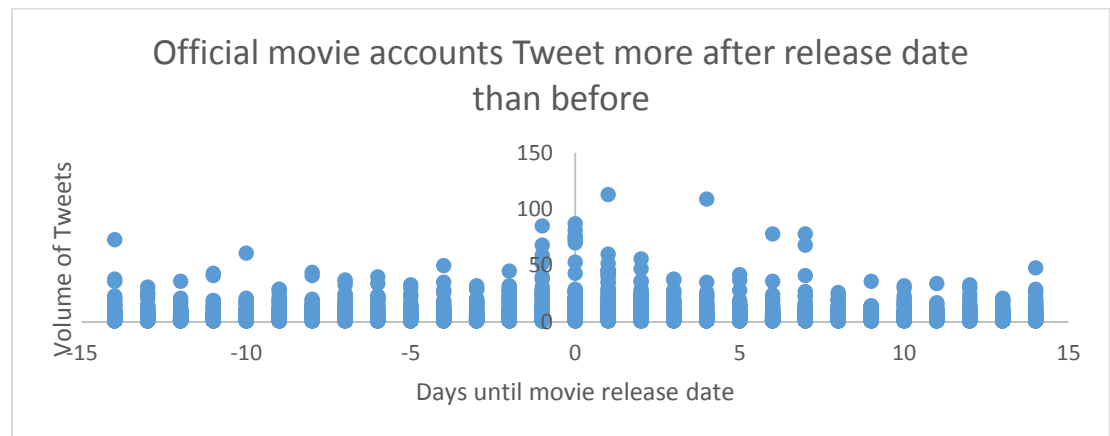
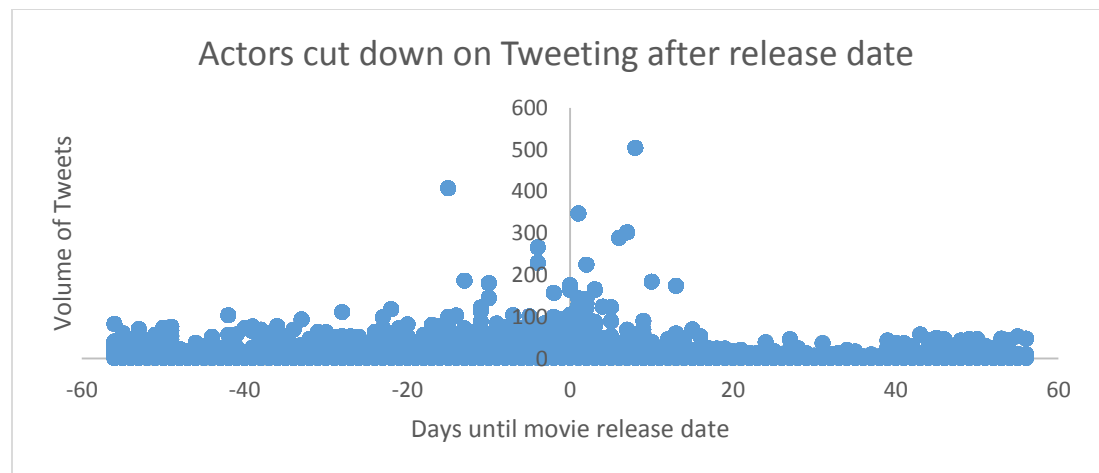


Figure 4.3



As can be seen by the graphs, movie accounts Tweet somewhat randomly, with the exception of the days near the release date, where these accounts seem to Tweet more. Time-series analysis shows that actor accounts also Tweet more after a movie's release date than before. Actors appear to be fairly active on Twitter prior to their movies' release date, and even more so on their movies' release date. However, unlike official movie accounts, shortly after the release date, there is a sharp drop in the amount of Tweets per day by these actors. To confirm, like the volume of user-generated content in relation to time, regression analysis statistically concludes that actor accounts Tweeted more as a release date approached and Tweeted less as the release date passed.

The primary reason behind firms' attempts to influence user-generated content regarding their product is the belief that positive word of mouth marketing will lead to an increase in ticket sales. In the next section I will examine if such a notion is correct, and if so, if firms are capable of influencing an increase in ticket sales.

V. RESULTS

The movie industry puts value on user-generated content primarily because of its influence on potential moviegoers' opinions. This being said, whether or not user-generated content affects the movie industry is unimportant if the movie industry cannot do anything to influence user-generated content. For this reason, my results section is divided into two sections (1) user-generated content's relationship with box-office revenue and (2) firm-generated content's relationship with user-generated content.

User-generated content's relationship with box-office revenue

Regression analysis was conducted to examine the relationship of user-generated content and box-office results. Along with measuring user-generated content, variables including genre, MPAA (parental) ratings, IMDb ratings, budget, actor/director prestige and week were measured to isolate user-generated content's effects. The full variable names, coefficients, t-values and p-values of each of the variables for Equation 1 are listed in Table 5.1.

Table 5.1

Estimation of results regarding the modeling of box-office revenue in Equation 1

Variable	Full Variable Name	Coefficient	t-value	p-value
<i>UGC</i>	Volume of User-generated Content	0.376	8.27	2.65*10⁻¹⁶
<i>Pos UGC</i>	Percent of Positive User-generated Content	0.028	0.06	.953
<i>Neg UGC</i>	Percent of Negative User-generated Content	-2.076*	1.81*	.070*
<i>ln(budg)</i>	Budget	0.445	5.38	8.34*10⁻⁸
<i>Dram</i>	Drama	-0.644	-2.43	.015

<i>Comd</i>	Comedy	0.347	1.29	.198
<i>Horr</i>	Horror	-0.304	-0.97	.334
<i>Acti</i>	Action	-0.073	-0.27	.789
<i>Dir</i>	Director	0.426	2.61	.105
<i>Act</i>	Actor	-0.079	-0.48	-.399
<i>GPG</i>	MPAA rating G or PG	1.284	5.06	4.64*10⁻⁷
<i>PG13</i>	MPAA rating PG-13	0.731	4.30	1.79*10⁻⁵
<i>IMDb</i>	IMDb rating	1.035	12.24	4.77*10⁻³³
<i>t</i>	Week	-0.621	16.45	1.79*10⁻⁵

Note. Indicator variables GPG and PG13 both equal 0 if a movie is rated R or unrated. Indicator variables *Dram*, *Comd*, *Horr* & *Acti* equal 0 if a movie is classified as an adventure film. Director and actor prestige is classified by whether an actor or director has been nominated for an Oscar. If *Dir* or *Act* equals 1, it means a director or actor has been nominated for an Oscar. **Bold text signifies $p < .05$** ; ‘*’ signifies $p < .1$.

When examining the impact of user-generated Tweets the week prior to a movie’s release, it can be deduced the volume of user-generated Tweets prior to a movie’s release date drives box-office sales. When examining the percentage of positive Tweets’ effect on box-office sales, no conclusion can be drawn due to the low t-value that accompanies the coefficient. This being said, the positive coefficient is produced as expected. On the other hand, the percentage of negative Tweets produced by the regression analysis is marginally significant and suggests that a higher percentage of negative Tweets may hurt box-office sales. These results are no surprise, reading positive and frequent discussions of a movie would suggest to potential moviegoers that the movie being discussed is worth watching. Negative or infrequent discussion would do the exact opposite.

Aligned with the literature, box-office revenue goes down over time after a movie’s release date (Ma, Singh & Smith, 2014). When examining MPAA ratings, a G or

PG rated movie drives more ticket sales than a PG-13 rated movie and a PG-13 rated movie more than a rated R movie. This is not surprising, as a more restrictive MPAA ratings reduce the number of viewers able to watch a movie. Furthermore, a more restrictive MPAA rating does not seem to give any incentive for potential moviegoers to watch a movie. Higher IMDb ratings also seem to help movies generate box-office revenue, which also makes intuitive sense, since higher IMDb ratings reflect movie quality. Finally, also as expected, higher movie budgets indicate stronger box-office revenue, possibly because a higher budget allows firms to recruit better talent and pay for higher quality movie production. Regarding genres, drama films fared the worst and were the only statistically significant genre in the regression analysis. The entire regression analysis contained an R^2 of 43.1%, indicating that the model explained 43.1% of variability of the response data.

As shown by user-generated content's effects on box-office sales, firms should be mindful of user-generated content. To increase the amount of Twitter conversation regarding their movie, films and their actors may attempt to direct conversation through their own official and personal accounts. For this reason, I examine firm-generated Tweet's relationship with user-generated Tweets. Specifically, I examine official movie Twitter accounts, these movies' actors' personal Twitter accounts and potential synergy between the interactions of these two types of accounts.

Firm-generated content's relationship with user-generated content

Movies

Examining firm-generated content's relationship with user-generated content fills a significant void in the literature and connects firm-generated content to box-office

sales. Using a methodology similar to the one that measured user-generated content's effects on box-office sales, a regression analysis was used to assess firm-generated content's relationship with user-generated content. Like Equation 1, the budget, IMDB sales and time³ were used to isolate firm-generated content's effects. Furthermore, indicator variables were used to control for genre and MPAA ratings. Unlike Equation 1, valence was not measured, as it was deemed fair to assume all firm-generated content has the intention of affecting the conversation positively. The full variable names, coefficients, t-values and p-values of each of the variables are listed in Table 5.2.

Table 5.2

Estimation of results regarding the modeling of user-generated content in Equation 2

Variable	Full Variable Name	Coefficient	t-value	p-value
<i>FGC</i>	Volume of Tweets from Movie Accounts	0.015	10.43	2.93*10⁻²⁵
<i>Budg</i>	Budget	1.09E-08	21.39	1.08*10⁻⁹⁷
<i>Dram</i>	Drama	0.421	5.43	5.86*10⁻⁰⁸
<i>Comd</i>	Comedy	0.473	6.10	1.13*10⁻⁰⁹
<i>Horr</i>	Horror	0.169*	1.85*	.064*
<i>Acti</i>	Action	0.332	4.05	5.24*10⁻⁵
<i>Dir</i>	Director	-0.299	-6.33	2.64*10⁻¹⁰
<i>Act</i>	Actor	-0.336	-6.96	3.69*10⁻¹²
<i>GPG</i>	MPAA rating G or PG	0.311	4.48	7.48*10⁻⁰⁶
<i>PG13</i>	MPAA rating PG-13	0.469	9.11	1.11*10⁻¹⁹
<i>IMDb</i>	IMDb rating	0.156	6.16	7.76*10⁻¹⁰

³ It is important to note that in order to lag time by 1 day, time is measured in units of days.

<i>t</i>	Day	-0.058	-38.64	3.5*10⁻²⁹⁰
----------	-----	---------------	---------------	------------------------------

Note. Indicator variables GPG and PG13 both equal 0 if a movie is rated R or unrated. Indicator variables *Dram*, *Comd*, *Horr* & *Acti* equal 0 if a movie is classified as an adventure film. Director and actor prestige is classified by whether an actor or director has been nominated for an Oscar. If *Dir* or *Act* equals 1, it means a director or actor has been nominated for an Oscar. **Bold text signifies $p < .05$** . ‘*’ signifies $p < .1$.

As hypothesized, the volume of content generated by firm accounts has a statistically significant and positive effect on the volume of user-generated content. PG-13 rated movies drive user-generated content more than G/PG rated movies and G/PG rated movies drive UGC-content more than rated-R movies. Twitter conversation being mostly driven by rated PG-13 movies is no surprise, since the typical Twitter user is old enough to watch a rated PG-13 movie, but not necessarily an R-rated movie. (Lenhart, 2015). Higher IMDb movie ratings and higher budgets seem to help user-generated content, which is not surprising since high movie ratings may allow for more publicity and higher budgets may allow for more marketing. As seen in the time series analysis, user-generated content goes down with time after a movie’s release. All genres were marked to have statistically significant effects on user-generated content in the following order: (1) Comedy/Romance, (2) Drama, (3) Action, (4) Horror and (5) Adventure. Interestingly, there is a negative coefficient associated with director and actor prestige in regards to user-generated content. This can be attributed to three possible phenomena: (1) movies without prestigious directors and actors leverage social media to advertise more, and therefore drive more Twitter conversation, (2) un-awarded directors and actors who are active on Twitter try to increase their brand value and remain competitive through regular interaction with potential moviegoers and/or (3) the potential mismatch between Twitter users, who are relatively young, and the target audience for Oscar-caliber movies, which consist of relatively older moviegoers.

Actors

The effects of actors' Tweets on user-generated Tweets were assessed using a methodology nearly identical to the one used to measure the effects of Tweets from movie-accounts. The full variable names, coefficients, t-values and p-values of each of the variables are listed in Table 5.3.

Table 5.3

Estimation of results regarding the modeling of user-generated content in Equation 3

Variable	Full Variable Name	Coefficient	t-value	p-value
<i>EGC</i>	Volume of Tweets from Actor Accounts	0.041	8.56	1.53*10⁻¹⁷
<i>ln(budg)</i>	Budget	4.44E-09	4.80	1.61*10⁻⁶
<i>Dram</i>	Drama	-0.055	-0.42	.67
<i>Comd</i>	Comedy	1.029	8.50	2.51*10⁻¹⁷
<i>Horr</i>	Horror	0.773	5.48	4.44*10⁻⁸
<i>Acti</i>	Action	0.239*	1.85*	.064*
<i>Dir</i>	Director	-0.605	-7.59	3.74*10⁻¹⁴
<i>Act</i>	Actor	-0.326	-4.37	1.26*10⁻⁵
<i>GPG</i>	MPAA rating G or PG	0.250	2.27	.023
<i>PG13</i>	MPAA rating PG-13	1.043	12.94	1.23*10⁻³⁷
<i>IMDb</i>	IMDb rating	0.633	16.19	2.79*10⁻⁵⁷
<i>t</i>	Day	-0.070	-33.70	3.5*10⁻²⁹⁰

Note. Indicator variables GPG and PG13 both equal 0 if a movie is rated R or unrated. Indicator variables *Dram*, *Comd*, *Horr* & *Acti* equal 0 if a movie is classified as an adventure film. Director and actor prestige is classified by whether an actor or director has been nominated for an Oscar. If *Dir* or *Act* equals 1, it means a director or actor has been nominated for an Oscar. **Bold text signifies p < .05.** '*' signifies p < .1.

Not surprisingly, compared to Formula 2, coefficients that were positive remained positive and those that were negative remained negative. Regarding MPAA ratings, relative variable strength remained the same, yet were re-ordered for genres.⁴ Like the movie accounts, employee-generated content was shown to drive user-generated content. Again, negative coefficients attributed to directors and actors who have received an Oscar nomination suggest that those with such nominations are less successful in driving user-generated content.

Movies and Actors

When examining a regression model that considers both movie account Tweets and employee account Tweets' interactions with user-generated Tweets, both these variables are deemed insignificant. However, because both categories of Tweets were deemed statistically significant in the two earlier models, their insignificance in this model could be attributed to the possibility that the variable FGC*EGC captured their effects on user-generated content. The regression attributed a positive coefficient to this variable, implying that employee and firm-generated content's interaction produces synergy to increase user-generated content. Like all other models built, distance has a negative effect on user-generated content, along with director and actor prestige. Also, like all other models, higher IMDb ratings and a higher budget fair well for user-generated content. The full variable names, coefficients, t-values and p-values of each of the variables are listed in Table 5.4.

⁴ For example, horror/thriller movies had a larger coefficient than action movies when examining employee generated content, but the opposite was observed when examining just movie account generated content.

Table 5.4

Estimation of results regarding the modeling of user-generated content in Equation 4

Variable	Full Variable Name	Coefficient	t-value	p-value
<i>FGC</i>	Volume of Tweets from Movie Account	1.970	1.04	.298
<i>EGC</i>	Volume of Tweets from Actor Account	4.754	1.42	.154
<i>EGC*FGC</i>	Movie and Actor Account Multiplier	.560	5.33	1.18*10⁻⁷
<i>ln(budg)</i>	Budget	0	1.45	.147
<i>Dram</i>	Drama	-274.919	-3.13	.00177
<i>Comd</i>	Comedy	-137.672	-1.62	.106
<i>Horr</i>	Horror	-119.021	-1.21	.225
<i>Acti</i>	Action	-87.1024	-.70	.484
<i>Dir</i>	Director	-161.182	-2.88	.004
<i>Act</i>	Actor	56.208	1.12	.264
<i>GPG</i>	MPAA rating G or PG	-25.052	-.36	.719
<i>PG13</i>	MPAA rating PG-13	343.136	6.78	2.04*10⁻¹¹
<i>IMDb</i>	IMDb rating	15.206	.58	.564
<i>t</i>	Day	-7.384	-7.38	3.12*10⁻¹³

Note. Indicator variables GPG and PG13 both equal 0 if a movie is rated R or unrated. Indicator variables *Dram*, *Comd*, *Horr* & *Acti* equal 0 if a movie is classified as an adventure film. Director and actor prestige is classified by whether an actor or director has been nominated for an Oscar. If *Dir* or *Act* equals 1, it means a director or actor has been nominated for an Oscar. **Bold text signifies p < .05.**

VI. CONCLUSION

For years, movie production firms have advertised their movies through billboard signs, posters and trailers – despite the fact that word-of-mouth marketing drives movie success. Those who enjoy a particular movie are quick to publicly praise it, while those who dislike a movie usually have no hesitation to lambaste the movie. Nowadays, such conversation is often shared on social media, allowing marketing teams to inject themselves into the conversation regarding their movie. However, a question worth asking is would an official movie account or actor account Tweeting drive potential moviegoer Tweets? And if so, would an increase in user Tweets drive box-office sales? Currently, the effects of firm-generated content on user-generated content have not been examined in the literature. This research fills this void in the literature by answering the question, “How does firm-generated content influence user-generated content and box-office sales?”

To answer this research question, data for 224 movies for two years was collected. 17,014,979 user-generated Tweets and 137,312 firm-generated Tweets were also examined for analysis. Regression analysis is conducted in this study to establish a relationship between user-generated content and box-office sales. Afterwards, regression analysis is conducted to examine the effects of firm-generated content on user-generated content.

Consistent with previous literature, I first find that the volume of user-generated content is shown to drive box-office sales (Asur & Huberman, 2010; Tirunillai & Tellis, 2012; Wong & Chiang, 2012). Also, the analysis of firm-generated content’s relationship with user-generated content shows that official movie account Tweets and personal actor

account Tweets drive user-generated content. These two findings together indicate that the firm-generated content drives box-office sales indirectly through its impact on user-generated content. Furthermore, my results show that the interaction between movie accounts and actor accounts have a positive effect on user-generated content.

The findings in this paper can make contributions to the movie marketing fields. The managerial implication that movie accounts and actor accounts' interactions produce a synergy which drives user-generated content should be taken advantage of by movie marketing teams. Oftentimes, these accounts run independently from one another and craft messages only to appeal to their followers – intentionally, not interacting with other accounts to avoid spreading a diluted message. Also, the interactions between movie accounts, actor accounts and potential moviegoer accounts are largely made up of retweets. In fact, retweets take up a large portion of movie accounts' Tweets and the plurality of actor Tweets, as can be seen in Appendix B and Appendix C. The findings in this study do not state that direct user interaction or Retweets are ineffective; however, there is quantifiable strength to the interactions between movie and actor accounts. Interestingly enough, actor and director prestige have negative effects regarding firm-generated content's relationship to user-generated content. Two of the hypotheses for this unexpected phenomenon include that less prestigious actors often try to compensate for their lack of recognition with direct interaction with fans on social media and that the best practices usually assumed by more prestigious actors may be having the opposite intended effect. Regardless of the reason, it would seem that prestigious actors would be recommended to mimic less prestigious actors' practices on social media to drive user-generated content. As this study suggests, such interactions would be beneficial to an

actor, regardless of prestige, since they directly boost user-generated content discussing the movie, indirectly boosting box-office sales.

This research has several limitations, which open directions for future research. In this study, I only focus on the volume and valence of firm- and user-generated content in analysis. Future research could investigate a richer set of activities among users and firms on social media. For example, movie accounts, actor accounts and user accounts on Twitter can interact with each other through replies, retweets and tagging. An examination of the dynamics of different types of interactions and the effects of these interactions can be a fruitful direction for future research. Furthermore, a generalization of our study to other industries will be extremely valuable. This thesis does not study the direct relationship between firm-generated content and box-office sales. Thus, future research could examine if and how firm-generated content directly drives box-office sales.

APPENDIX A

Summary Statistics of Movie Sample

	Mean	Median	Standard Deviation	Maximum	Minimum
Box-Office Sales (Week 1)	\$21,512,539	\$8,521,412	\$34,995,885	\$296,211,625	\$7,830
Box-Office Sales (Week 2)	\$12,073,580	\$4,910,119	\$17,950,889	\$149,629,150	0
Box-Office Sales (Week 3)	\$6,912,531	\$2,929,990	\$10,224,945	\$81,460,150	0
Budget	\$44,194,857	\$52,458,637	\$22,000,000	\$250,000,000	\$200,000
IMDb Rating	6.56	0.98	6.60	8.70	1.60

	Action	Adventure	Comedy/Romance	Drama	Horror/Thriller
Total Movies	28	24	60	73	39

	G&PG	PG-13	R/Un-rated
Total Movies	30	89	105

	Nominated for Oscar	No Nomination
Director	70	154
Actor	121	103

APPENDIX B

Breakdown of Movie Account Tweets

	Mean	Median	Standard Deviation	Maximum	Minimum
Volume of All Tweets/day (Week prior to release)	2,557	2,533	836	3,844	1,440
Volume of All Tweets/Day (Week of Release)	2,764	2,423	1,368	5,086	1,450
Volume of All Tweets/Day (Week 2 after Release)	1,065	956	362	1,583	690
Volume of Replies/day (Week prior to release)	291	275	121	488	109
Volume of Replies Tweets/Day (Week of Release)	405	332	197	680	157
Volume of Replies Tweets/Day (Week 2 after Release)	115	108	36	181	79
Volume of Retweets Tweets/day (Week prior to release)	1,253	1,170	519	2,017	674
Volume of Retweets Tweets/Day (Week of Release)	1,549	1,278	941	2,987	688
Volume of Retweets Tweets/Day (Week 2 after Release)	448	405	115	661	352
Retweets per Tweet (Week prior to release)	106	9	999	70,867	9
Retweets per Tweet (Week of Release)	113	6	1,634	148,297	6
Retweets per Tweet (Week 2 after Release)	63	5	819	67,115	5
Favorites per Tweet (Week prior to release)	55	0	399	23,620	0
Favorites per Tweet (Week of Release)	39	0	270	8,322	0
Favorites per Tweet (Week 2 after Release)	42	0	266	9,163	0

APPENDIX C

Breakdown of Actor Account Tweets

	Mean	Median	Standard Deviation	Maximum	Minimum
Volume of All Tweets/day (Week prior to release)	795	703	251	1243	542
Volume of All Tweets/Day (Week of Release)	972	764	411	1659	618
Volume of All Tweets/Day (Week 2 after Release)	531	498	121	784	427
Volume of Replies/day (Week prior to release)	190	170	71	296	108
Volume of Replies Tweets/Day (Week of Release)	211	195	76	309	123
Volume of Replies Tweets/Day (Week 2 after Release)	138	123	36	207	104
Volume of Retweets Tweets/day (Week prior to release)	300	260	118	508	176
Volume of Retweets Tweets/Day (Week of Release)	419	291	235	792	243
Volume of Retweets Tweets/Day (Week 2 after Release)	186	168	61	306	131
Retweets per Tweet (Week prior to release)	852	16	44,745	3,336,761	0
Retweets per Tweet (Week of Release)	239	10	2,029	120,441	0
Retweets per Tweet (Week 2 after Release)	213	11	970	30,035	0
Favorites per Tweet (Week prior to release)	356	5	1,979	61,263	0
Favorites per Tweet (Week of Release)	322	2	2,561	155,273	0
Favorites per Tweet (Week 2 after Release)	300	6	1,142	19,835	0

APPENDIX D

Breakdown of User Account Tweets

	Mean	Median	Standard Deviation	Maximum	Minimum
Volume of Tweets/day (Week prior to release)	439	167	825	11,725	0
Volume of Tweets/Day (Week of Release)	978	343	2,082	35,681	0
Volume of Tweets/Day (Week 2 after Release)	1,971	518	4,926	77,601	0
Volume of Positive Tweets/day (Week prior to release)	136	45	273	4,493	0
Volume of Positive Tweets/Day (Week of Release)	310	94	738	13,308	0
Volume of Positive Tweets/Day (Week 2 after Release)	822	176	2,135	36,522	0
Volume of Negative Tweets/day (Week prior to release)	20	5	73	1,757	0
Volume of Negative Tweets/Day (Week of Release)	36	10	121	3,390	0
Volume of Negative Tweets/Day (Week 2 after Release)	151	25	584	12,680	0

REFERENCES

- Asur, S., & Huberman, B. (2010, August). Predicting the future with social media. *Web Intelligence and Intelligent Agent Technology* (pp. 492-499). Singapore. doi: 10.1109/WI-IAT.2010.63
- Chintagunta, P. K., Gopinath, S., & Venkataraman, S. (2010). The effects of online user reviews on movie box office performance: Accounting for sequential rollout and aggregation across local markets. *Marketing Science*, 29(5), 944-957. doi: 10.1287/mksc.1100.0572
- Gossieaux, F., & Moran, E. (2010). *The hyper-social organization: Eclipse your competition by leveraging social media*. New York City, New York: McGraw Hill Professional. doi: 10.1108/07363761211206438
- Goldfarb, A., & Tucker, C. (2012). Privacy and innovation. *Innovation Policy and the Economy*, 12, 65-89. doi: 10.3386/w17124
- Kumar, A., Bezawada, R., Rishika, R., Janakiraman, R., & Kannan, P. K. (2016). From social to sale: The effects of firm-generated content in social media on customer behavior. *Journal of Marketing*, 80(1), 7-25. doi: 10.1509/jm.14.0249
- Lenhart, A. (2015). Teens, Social Media, and Technology Overview 2015. Pew Research Center.
- Liu, A., Mazumdar, T., & Li, B. (2014). Counterfactual decomposition of movie star effects with star selection. *Management Science*, 61(7), 1704-1721. doi:10.1287/mnsc.2014.1923
- Ma, L., Montgomery, A. L., Singh, P. V., & Smith, M. D. (2014). An empirical analysis of the impact of pre-release movie piracy on box office revenue. *Information Systems Research*, 25(3), 590-603.
- Mayzlin, D. (2006). Promotional chat on the internet. *Marketing Science*, 25(2), 155-163. doi: 10.1287/mksc.1050.0137
- Miller, A. R., & Tucker, C. (2013). Active social media management: The case of health care. *Information Systems Research*, 24(1), 52-70. doi: 10.1287/isre.1120.0466
- PricewaterHouseCoopers LLP. (2015). Filmed entertainment: Key insights at a glance. Retrieved from <http://www.pwc.com/gx/en/industries/entertainment-media/outlook/segment-insights/internet-advertising.html>
- Rishika, R., Kumar, A., Janakiraman, R., & Bezawada, R. (2013). The effect of customers' social media participation on customer visit frequency and

- profitability: An empirical investigation. *Information Systems Research*, 24(1), 108-127. doi: 10.1287/isre.1120.0460
- Shahbaznezhad, H., & Tripathi, A. (2015). Customer and firm interactions on social media. *PAICS 2015 Proceedings*. doi: 10.2139/ssrn.2660685
- Shriver, S. K., Nair, H. S., & Hofstetter, R. (2013). Social ties and user-generated content: Evidence from an online social network. *Management Science*, 59(6), 1425-1443. doi: 10.1287/mnsc.1110.1648
- Tirunillai, S., & Tellis, G. J. (2012). Does chatter really matter? Dynamics of user-generated content and stock performance. *Marketing Science*, 31(2), 198-215. doi: 10.1287/mksc.1110.0682
- Wong, F. M. F., Sen, S., & Chiang, M. (2012, August). Why watching movie tweets won't tell the whole story? *Proceedings of the 2012 ACM Workshop on Workshop on Online Social Networks* (pp. 61-66). Helsinki, Finland. doi: 10.1145/2342549.2342564